

Epistemic Border Control as Proxy Regulation Under Partial Observability

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Abstract

Informal regulatory systems and institutional feedback controllers often attempt to govern targets — claim quality, truth-tracking, infrastructure burden, epistemic hygiene — under partial observability. When no real-time sensor on the target exists, or its signal-to-noise is dominated by a distinct proxy variable, the controller is structurally driven to substitute the proxy for the nominal target — strictly in the unobservable case, practically in the weakly-observable case as observability conditioning deteriorates. The rhetoric of regulation remains fixed on the nominal target while the actual controlled variable becomes the proxy. This paper isolates the substitution as a consequence of observability asymmetry, offers a finite-horizon observability-Gramian diagnostic for the substitution magnitude in the simulated regime, and operationalizes both via a single-agent Kalman-LQR simulation. The substitution does not require corrupt intent, gamed metrics, or slow drift: sincere regulators with a correctly-specified cost function and a correctly-specified plant model produce the same substitution under the same sensor geometry. The result is sibling to Paper 24's [2] aggregation-layer masking by an algebraic argument — aggregation reduces variance as $O(\sigma^2/N)$ but does not rotate the observability subspace — and shares closed-loop substitution geometry with Paper 23's [1] identifiability-masking apparatus. Informal discourse moderation is the worked case in §7; the contribution is a control-geometry result that applies wherever a controller with a fixed sensor map confronts a target outside its observable subspace.

1. Thesis

The target is unsensed. The proxy is sensed. The controller acts through the sensed channel. Substitution is therefore not corruption, laziness, or Goodharting. It is the control consequence of observability asymmetry.

The load-bearing claim has two regimes:

- **Strict unobservability.** When target values produce identical observation trajectories, the controller cannot regulate what its sensorium does not admit. This is the impossibility result of Theorem 1.
- **Weak observability / ill-conditioning.** When the target enters the observation map but is dominated by the proxy, target tracking is in principle possible but practically collapses into proxy tracking as observability conditioning deteriorates. This is the regime simulated in §4.

In both regimes: when the nominal target V is not effectively observable and a proxy V' is, sincerity changes the moral status of substitution but not its control geometry. Sincere regulators facing the same sensor set produce the same substitution. The mechanism is structural; the action is agential — but the locus of agency is narrow. The structural constraint determines what the controller can distinguish; agency remains at the consequence boundary — whether the actor treats proxy evidence as grounds for inquiry,

advisory caution, or target-binding action. Responsibility is not for failing to observe the unobservable; responsibility is for treating proxy evidence as target authority.

This distinguishes the present claim from at least three adjacent positions, each developed in §6:

- **vs Goodhart.** Goodhart’s Law says a metric becomes a target — a contingent corruption that better metrics or more disciplined measurement could in principle avoid. The present claim is sharper: when the underlying target has no real-time sensor at all, no amount of sincere intent produces target-tracking. The substitution is necessary, not contingent.
- **vs performative prediction.** Performative prediction (Perdomo et al. [4]) addresses the case where a model’s output changes the data distribution it later trains on. The present claim does not require performativity: the substitution arises *before* the controller’s actions perturb the environment, by virtue of the sensor map alone.
- **vs platform-moderation trade-off literature.** Existing moderation literature addresses domain-specific balancing under explicit policy regimes. The present claim is a general control-geometry result that applies wherever a controller with a fixed sensor map confronts a target lying outside the observable subspace.

The constructive consequence — what an architecture must do to prevent the substitution rather than merely apologize for it — is sketched in §8.

Classical observability explains why the hidden target cannot be recovered from the proxy channel alone. The admissibility claim here is that a controller acting only on proxy-derived feedback cannot legitimately bind its action to the target it cannot observe. This is the partial-observability instance of the series’ broader authority-binding refusal: a mechanism may be well formed at one layer while lacking standing to bind, mutate, or authorize at another. The control-theoretic result is classical; the contribution is the structural reading of classical observability as an admissibility limit on proxy-derived authority. The theorem below is not presented as a novel control-theoretic result; its work is interpretive — converting an observability fact into an authority constraint.

2. Model

2.1 State, observation, and control

Let $x_t \in \mathbb{R}^n$ be the plant state. Let the nominal target be a scalar $V_t = q^\top x_t$ for some weight vector $q \in \mathbb{R}^n$. Let the controller’s available observations be $y_t = C_{\text{obs}} x_t + v_t$, with $C_{\text{obs}} \in \mathbb{R}^{m \times n}$ and noise v_t . Let the controller policy depend only on the observable trajectory: $u_t = K(y_0, \dots, y_t)$.

For concreteness throughout the paper we will work with a two-latent decomposition $x_t = [T_t, C_t]^\top$, where T_t is the slow, expensive-to-measure target component (object-level claim quality, truth-tracking, infrastructure burden) and C_t is the fast, cheap-to-measure proxy component (reputational contamination, stylistic genre signal, coalition-adjacency risk). The cost function is

$$J = \mathbb{E} \left[q_T T_t^2 + q_C C_t^2 + \lambda u_t^2 \right],$$

with the rhetoric-stays-fixed-on- V axiom encoded as $q_C = 0$: the controller’s *stated* objective is concentrated on T . The plant matrix A , control matrix $B = [b_T, b_C]^\top$, and observation map $C_{\text{obs}} = [\alpha_T, \alpha_C]$ are otherwise standard.

2.2 The observability asymmetry

The load-bearing structural feature: C_{obs} loads heavily and quickly on C_t and only weakly and slowly on T_t . In the discourse-moderation instance: rhetoric, source cues, coalition signals, and stylistic genre are observable in real time; the underlying truth claim is not.

Equivalently, the finite-horizon observability matrix

$$O_T = \begin{bmatrix} C_{\text{obs}} \\ C_{\text{obs}}A \\ \vdots \\ C_{\text{obs}}A^{T-1} \end{bmatrix}$$

has its least-observable singular direction $v_{\min}(O_T)$ aligned with the cost direction q (the T -axis in the two-latent case) when α_T/α_C is small.

2.3 What the controller can do

The controller's policy depends on $y_{0:t}$ via a state estimator (e.g. a Kalman filter) and a feedback law (e.g. LQR). Two states x, x' that produce identical observation trajectories $C_{\text{obs}}A^k x = C_{\text{obs}}A^k x'$ for all $k < T$ are observationally indistinguishable to any policy with horizon $\leq T$, regardless of the controller's *cost function* on those states. This is the structural fact the rest of the paper exploits.

3. Substitution-forcing theorem

We state the substitution claim in two forms: a static algebraic version (Theorem 1) for the strict-unobservability case, and a regime-specific finite-horizon Gramian diagnostic (Diagnostic 1) that tracks the substitution magnitude in the weakly-observable / simulated regime.

3.1 Theorem 1 (static observability-asymmetry substitution)

Theorem 1 (noiseless case). *Let the observation map be deterministic ($v_t \equiv 0$). Let $x, x' \in \mathbb{R}^n$ be plant states with $C_{\text{obs}}A^k x = C_{\text{obs}}A^k x'$ for all $k = 0, \dots, T-1$, but $q^\top x \neq q^\top x'$ (i.e., distinct values of the nominal target V). Then any controller whose policy depends only on $\{y_0, \dots, y_{T-1}\}$ assigns the same control action sequence to x and x' . Consequently, no such controller can condition its action on the difference in V between these states. Any effect on V is therefore not target-specific regulation of V through observation, but an undifferentiated action over observationally aliased states. The regulable component of the policy is restricted to what the observation trajectory makes distinguishable.*

Proof. By construction the observation trajectories are identical: $y_k = C_{\text{obs}}A^k x = C_{\text{obs}}A^k x'$ for all $k < T$. Any policy $K(y_0, \dots, y_{T-1})$ depends on the trajectory only through these observations and so assigns the same $u_{0:T-1}$ to both initial conditions. The closed-loop trajectory of $V_t = q^\top x_t$ then differs between x and x' — by an amount that propagates the initial-condition difference $q^\top(x - x')$ through the closed loop — without the controller having any policy degree of freedom to track that difference. The controller can therefore regulate only distinctions preserved by the observation trajectory; any action on V is mediated through the observable quotient — what we informally call the proxy V' — rather than through target-specific access to V . ■

Remark (classical anchor). Theorem 1's structural fact — states identical to the controller across the observation trajectory are indistinguishable to any policy with horizon $\leq T$ — is the closed-loop reading of

Kalman’s observability decomposition (Sontag [8], ch. 6). The control-theoretic content is classical; the contribution here is the substitution-forcing reading of that classical fact, prior to any cost-function or sincerity considerations.

Remark (noisy case). Adding observation noise v_t does not lift indistinguishability in the strict case: the noiseless argument captures the structural fact that observation geometry forecloses regulation when trajectories are exactly identical. In the weakly-observable case — where target distinguishability is in principle present but ill-conditioned — noise does not produce strict impossibility but converts the geometry into estimation-conditioning / SNR dependence; near-identical observable trajectories produce near-identical control actions, and the substitution magnitude is then tracked, in the simulated regime, by Diagnostic 1 below, with the noise floor entering through the Kalman posterior covariance. The deterministic theorem supplies the strict indistinguishability boundary; the noisy and weakly-observable cases are treated by Diagnostic 1 and the simulation rather than by Theorem 1 itself.

The theorem says nothing about whether the controller’s stated cost weights the unobservable component of V . It says only that observation geometry already forecloses the regulation: a sincere controller with $q_C = 0$ and a correctly-specified cost is in exactly the same position as a corrupt controller that explicitly swapped V for V' .

3.2 Diagnostic 1 (Gramian-conditioning diagnostic)

This is a regime-specific diagnostic, not a proved theorem. Statement and rationale are explicitly limited to the simulated LTI / Kalman-LQR setting; the fully general claim is named below as open.

Diagnostic 1. *Let $W_o = O_T^\top O_T$ be the finite-horizon observability Gramian. For an LTI system with cost concentrated on a state direction q , the steady-state substitution index — the gap between the closed-loop tracking error in $V = q^\top x$ achieved by an ideal observer of V and the tracking error achieved by the available controller — is tracked, in the regime simulated in §4, by $|\langle v_{\min}(O_T), q \rangle|$ and by the conditioning of W_o along q . The substitution index increases as both quantities increase in this regime, independently of controller sincerity or model correctness.*

Diagnostic rationale. In the standard LTI estimation setting, the Kalman posterior covariance along the cost direction q is controlled by the observability conditioning in the q direction; in the finite-horizon / low-process-noise approximation this appears as $q^\top W_o^{-1} q$ at steady state, modulo process and observation noise (Anderson & Moore [9], ch. 4), where W_o is taken to be invertible on the relevant subspace, or under the usual regularized / noise-conditioned interpretation of the Kalman covariance when W_o is singular along $\ker(O_T)$. When q aligns with $v_{\min}(O_T)$, this quantity grows as the inverse of the smallest singular value $\sigma_{\min}(O_T)^2$. The LQR feedback law applied to the resulting estimate $\hat{x}$ propagates the posterior uncertainty into the closed-loop tracking error; in the simulated regime, the substitution magnitude is therefore tracked by $|\langle v_{\min}, q \rangle|$ and by $1/\sigma_{\min}(O_T)$. The bridge from finite-horizon observability Gramian to steady-state Kalman-LQR tracking error is regime-specific (LTI plant, low process noise, well-specified cost), not a general theorem. The fully general claim across all (A, B, Q, R) and noise structures is open. ■

The simulation in §4 traces the empirical scaling in this regime: $|\langle v_{\min}, e_T \rangle|$ varies from 0.9999 to 0.0439 as α_T/α_C varies from 0.01 to 10, and the substitution magnitude scales correspondingly across two orders of magnitude.

4. Simulation and Gramian bridge

4.1 Single-agent Kalman-LQR setup

A single-agent simulation (`paper25_substitution.py`; see `docs/formalization-index.md` for the companion-artifact crosswalk) instantiates the model of §2 with a correctly-specified Kalman-LQR controller. The two-latent process has T_t as a random walk and C_t as an AR(1) process with Poisson crank-shock innovations. Observation $y_t = \alpha_T T_t + \alpha_C C_t + v_t$. Control affects both latents asymmetrically: $b_C = -1.0$, $b_T = -0.05$. The cost function has $q_C = 0$ — the rhetoric-stays-fixed-on- V axiom is baked in at the cost function, not smuggled in through controller misspecification.

Scope of the simulation. This simulation illustrates one correctly-specified Kalman-LQR regime — fixed plant, fixed cost, Gaussian-with-Poisson-shock innovations, five seeds per parameter point, no confidence bounds reported. It is offered as a mechanism check, not as a universal scaling law across (A, B, Q, R) . The numbers below are representative of this sweep, not derived bounds.

4.2 Phase-transition sweep

Varying α_T/α_C from 0.01 to 10 (five seeds per ratio, 3000 steps each, 500-step burn-in) produces a power-law scaling of substitution magnitude. In the table below, $e_T = [1, 0]^\top$ denotes the unit vector in the target (T) direction.

α_T/α_C	$\sigma_{\min}(O_T)$	$e_T^\top W_o e_T$	$ \langle v_{\min}, e_T \rangle $	$T_{\text{rms}}^{\text{asym}}/T_{\text{rms}}^{\text{clean}}$
0.01	0.0226	0.002	0.9999	333
0.05	0.1127	0.050	0.9964	175
0.20	0.4278	0.80	0.9435	57
0.50	0.8364	5.0	0.7145	25
1.00	1.0499	20.0	0.4215	13
2.00	1.1248	80.0	0.2178	7.2
10.0	1.1509	2000	0.0439	1.9

Reading the table left-to-right, the T -axis rotates into and out of the observability null-space as the sensor weighting varies. At small α_T the least-observable direction $v_{\min}$ is essentially the T -axis itself (alignment ≈ 1); at large α_T the alignment has rotated away (≈ 0) and the ill-conditioned direction is now the C -axis. In this sweep, the substitution magnitude $T_{\text{rms}}^{\text{asym}}/T_{\text{rms}}^{\text{clean}}$ tracks the alignment column closely, with the ratio scaling roughly as $(\alpha_T/\alpha_C)^{-0.75}$ across the five seeds per parameter point. The qualitative ordering was stable across seeds; the table reports representative sweep values. The exponent is a fit to this sweep, not a derived law, and full variance reporting is deferred — per §4.1, the simulation is used as a mechanism check, not as an empirical scaling law.

4.3 Effort without effect

A second diagnostic isolates the rhetorical payload. At $\alpha_T/\alpha_C = 0.05$ in this representative sweep: $|U|_{\text{asym}} = 0.20$ vs $|U|_{\text{clean}} = 0.09$, with $T_{\text{rms}}^{\text{asym}} = 6.8$ vs $T_{\text{rms}}^{\text{clean}} = 0.04$. The asymmetric-sensor controller in this point is doing more than twice the control work for roughly 180× worse tracking. The work is not wasted; it is going into C -suppression. The controller’s subjective report — its cost function — remains “I am regulating T .” The actual closed-loop variable being regulated is C , because that is where the posterior signal lives.

This is the rhetoric-stays-fixed phenomenon stated in two numbers.

4.4 Mechanism confirmation

The LQR gain is $[-2.92, 0]$ across the entire sweep — the controller assigns zero weight to C in its cost function, as the $q_C = 0$ axiom requires. The substitution runs entirely through the Kalman filter: $\hat{T}$ is pulled around by C because $y = \alpha_T T + \alpha_C C$ and α_C dominates. LQR responds to the contaminated $\hat{T}$ via $B = [b_T, b_C]^\top$ with $|b_C| = 20|b_T|$, so the same control input mostly suppresses C . No misspecification, no wrong beliefs, no corrupted intent — in this regime, estimator geometry drives the observed substitution.

4.5 A second substitution channel: shock-statistics mismatch

Even at $\alpha_T/\alpha_C = 10$, $T_{\text{rms}}^{\text{asym}}/T_{\text{rms}}^{\text{clean}} \approx 1.9$, not unity. The remaining gap arises from the filter’s Gaussian model being blindsided by Poisson crank-shock innovations on C . A unit-amplitude shock ten times the one-step C -sigma arrives instantaneously; the filter, expecting smooth Gaussian C -innovations, misattributes part of the resulting y -jump to T . That drives $\hat{T}$ up, LQR applies u , and the substitution cascade fires — even though α_T is large enough that the smooth- C component would be well-separated.

This suggests a second sufficient channel, distinct from the sensor-angle argument that is the paper’s main contribution. In addition to observation-weight asymmetry ($\alpha_T \ll \alpha_C$), filter mis-specification with respect to the real state statistics on a channel coupled to y may produce substitution; the Gaussian filter’s blindness to Poisson jumps in this simulation is one instance. Diagnostic 1’s geometric condition is one sufficient channel for substitution in this regime; shock-model mismatch is a second, independent sufficient channel. The latter does not weaken the observability-Gramian diagnostic; it shows that observability asymmetry is not the only route by which proxy-coupled disturbances can enter the target estimate. The shock-statistics channel is flagged as future work and not developed further; the present paper restricts itself to the sensor-geometry channel, and treatment of filter-mis-specification substitution belongs to a separate object.

4.6 Bridge to Paper 23

Paper 23 [1] §3.3 case (ii) — the *measurement null-space masking* result — formalizes the case where a compensator H_t acts in the null space of the measurement map and is observationally invisible to first order over a horizon T . The present paper instantiates a different consequence of the same geometric object: the cost-targeted state direction q lies near $\ker(O_T)$, and the closed-loop control problem inherits ill-conditioning. Paper 23 case (ii) is the identity-axis manifestation (compensator invisibility); the present paper is the action-axis manifestation (regulation infeasibility). Both invoke $\ker(O_T)$ or its near-kernel and use it to state distinct closed-loop consequences. The present paper inherits Paper 23’s formal apparatus rather than reconstructing it, and theorem statements above cite §3.3 case (ii) directly.

5. Distinction from Paper 24

Paper 24 [2] names *aggregation-layer masking* as a structurally distinct failure mode of multi-agent episodic regulation. The present paper is the *spatial substitution* counterpart, sibling rather than nested. The sibling adjudication is algebraic, not empirical:

	Paper 24	This paper
Scale	Organizational governance	Discourse / community moderation (single or multi-agent)
Nominal objective	Plant state tracking via V_t	Truth-tracking via epistemic hygiene
Failure layer	Aggregation + witness-inclusion filter	Observability-driven target substitution
Effect on V_t	<i>Freezes</i> against updates	Gets silently <i>replaced</i> by V'_t
Architectural fix	Shape-preserving aggregation + open witness inclusion	(Open — see §8)

The substitution mechanism is structurally distinct from freeze. Freezing V is not the same as replacing V with V' .

Algebraic adjudication

The two structural facts of this section are recorded in the companion Lean as the **kernel-preservation lemma** (`ker_replicateRows_eq_ker`) and the **Gramian-scaling identity** (`replicateRows_transpose_mul1`); the paper-side crosswalk is `docs/formalization-index.md`.

Let N homogeneous agents observe the same latent state $x_t = [T_t, C_t]^\top$ through a common measurement map $C_{\text{obs}} = [\alpha_T, \alpha_C]$ with independent additive noise: $y_{i,t} = C_{\text{obs}}x_t + v_{i,t}$, $i = 1, \dots, N$. Stack the observations as $\tilde{y}_t = \tilde{C}x_t + \tilde{v}_t$ with $\tilde{C} = \mathbf{1}_N \otimes C_{\text{obs}}$. Then:

- The stacked finite-horizon observability matrix $O_T^{\text{stack}} = \mathbf{1}_N \otimes O_T$. Consequently $\ker(O_T^{\text{stack}}) = \ker(O_T)$.
- The least-observable eigenspace / subspace is unchanged; in the nondegenerate case (smallest singular value of O_T simple) this can be represented by the same $v_{\min}$ up to scale and sign.
- The stacked-observation Kalman filter has posterior variance $O(\sigma^2/N)$ — aggregation improves SNR — but the *direction* of residual uncertainty is unchanged.
- The LQR gain depends only on (A, B, Q, R) and is identical to the single-agent case. This invariance is restricted to the homogeneous replicated-measurement comparison; the heterogeneous case is treated under the scope condition below.

Therefore: Paper 24’s clean aggregation (shape-preserving, open witness inclusion) reduces observation noise but does not rotate the observability subspace. The measurement-noise-driven component of substitution can shrink like $O(1/\sqrt{N})$ under independent homogeneous observations, but the residual direction of uncertainty, the kernel of O_T , and any structural / model-mismatch channel — including the shock-statistics channel of §4.5 — remain unchanged. Paper 24’s sufficient condition for freeze-freedom is *not* sufficient for substitution-freedom. The mechanisms are independent.

The core line, in one sentence: *aggregation improves SNR; it does not rotate the observability subspace.*

Subspace, not vector. A technical caveat is worth making explicit. When the smallest singular value of O_T is degenerate, there is no distinguished “least observable direction” $v_{\min}$; the invariant claim of the algebraic argument is about the unobservable subspace, not a privileged vector. Concretely, homogeneous-witness replication scales the Gramian,

$$(\mathbf{1}_N \otimes O_T)^\top (\mathbf{1}_N \otimes O_T) = N \cdot O_T^\top O_T,$$

preserving its eigenspaces (and in particular its kernel) and multiplying its eigenvalues by N . Aggregation improves weight along already-observed directions; it does not create epistemic access to a direction outside the observation subspace. The formal statements are the Lean lemmas named at the head of this section.

Scope condition for the algebraic argument

Homogeneity of the measurement map across agents is load-bearing. Heterogeneous agents with genuinely different measurement maps $C_{\text{obs}}^{(i)}$ — for instance, a privileged witness with access to T through a different sensor channel — *can* rotate the effective observability subspace, and the sibling claim does not extend to that case without further argument. The heterogeneous-agent extension is a Paper 24/25 follow-on object — *what cohort-witness composition restores T -observability?* — not part of the sibling-vs-nested decision.

6. Distinction from Goodhart and from performative prediction

This section is the principal “not just X” firewall. The contribution depends on holding a sharper edge than the adjacent literatures supply.

The classical structural anchor for the strict-unobservability case is Francis and Wonham’s [10] *internal model principle*: exact regulation of a reference signal requires the controller to embed a model of that reference. The present paper uses this as a structural analogue rather than a direct theorem import: where no observation channel supports an internal model of the target, exact regulation of that target is unavailable. When the underlying target T has no real-time sensor, no internal model of T can be derived from the observable trajectory, so exact regulation of T is structurally impossible. The sub-sections below treat distinctions from contemporary adjacent literatures; the foundational anchor is fifty years older and structurally complete on its own terms. The paper’s contribution is not to refine the internal model principle but to read it normatively as an admissibility claim about proxy-derived authority.

6.1 vs Goodhart’s Law

Goodhart’s Law and its descendants describe a *contingent corruption*: when a measure becomes a target, optimization pressure tends to game the measure. The literature treats this as a problem amenable to better metrics, more robust measures, or counterfactual auditing.

The present claim is *upstream-framed*. When the underlying target has no real-time sensor at all — when T does not enter the observation map C_{obs} — no metric is being gamed, because the target was never available to optimize against in the first place. When the target enters but is dominated by the proxy, the controller can in principle track it but practically does not, because the posterior signal lives in the proxy direction. The controller is not corrupted; in the strict-unobservability case it is structurally barred from regulating T , and in the weakly-observable case it is practically pushed away from regulating T by ill-conditioning. Sincere intent does not save it.

The pithy form: Goodhart says *the measure is gamed*. The present claim says *the target never effectively entered the controller’s decision loop*. Put by direction: in Goodhart, the measure becomes the target; in observability-asymmetry substitution, the unsensed target never becomes a measure at all — the proxy

is what the controller can see, so it becomes what the controller can regulate. The two are compatible: observability-asymmetry substitution names a precondition that often precedes Goodhart-style proxy failure. Goodhart’s remedy — replacing the gamed metric with a less-gameable one — does not address the upstream condition where the target itself is unavailable or only weakly available. That can only be addressed by restoring or strengthening the T -channel, which is by hypothesis difficult or impossible.

Goodhart begins after a proxy has been selected and optimized; observability-asymmetry substitution names the prior condition in which the sensor map determines which proxy can become regulable at all.

Manheim and Garrabrant’s [3] classification of Goodhart variants — *regressional*, *extremal*, *causal*, *adversarial* — collectively covers the family of failures that occur when a proxy is selected, optimized, sampled, selected on, or adversarially exploited. Each variant assumes a regime in which a proxy has already been chosen and is being relied upon, and each describes a way in which reliance fails. Observability-asymmetry substitution sits *upstream* of these variants — particularly regressional Goodhart: before optimization pressure corrupts a chosen proxy, the sensor map may already determine which proxy is regulable in the first place. The two mechanisms are compatible. The contribution here is not to claim the four Goodhart variants do not apply; they describe the post-selection dynamics that follow once observability geometry has narrowed the regulable surface. The present paper isolates the upstream condition under which a proxy becomes the only available object.

6.2 vs performative prediction (Perdomo et al. [4])

Performative prediction names the phenomenon where a model’s predictions shape the data distribution it later trains on. The closed loop is between model output and environment response. The mathematical apparatus addresses convergence to *performatively stable* points and the gap between performatively stable and performatively optimal solutions.

The present claim does not require performativity. The substitution arises *before* the controller’s actions perturb the environment, by virtue of the sensor map alone. A controller observing a static distribution through a T -blind sensor still substitutes V' for V . Performative prediction is a separate (and stackable) failure mode; the present paper isolates the sensor-geometry channel.

6.3 vs feedback-loop classifications (Pagan et al. [5])

Pagan et al. give a five-type classification of feedback loops in automated decision-making — *individual*, *feature*, *outcome*, *ML-model*, *sampling* — and trace how each contributes to bias amplification under re-training. Their taxonomy is broad enough that the present mechanism could be located within it (loosely, a feature-loop instance), but they do not name observability-driven target substitution as a distinct failure mode with a distinct architectural remedy: their loops are about how proxies get *amplified* once selected, not about how the structural unavailability of a real-time target sensor *forces* a proxy in the first place. The present paper is one entry their classification leaves implicit, given a closed-loop formal characterization.

6.4 vs recommender-system control models (Sprenger et al. [6])

Sprenger et al. give a closed-loop control formalization of recommender systems acting on opinion dynamics in social networks, with engagement maximization as the controller objective and full state visibility on the network. The present paper shares the control-theoretic posture but operates on a different surface and isolates a different pathology: target substitution forced by sensor-map asymmetry, rather than equilibrium drift under engagement-maximizing closed loops with observable state. Their model is closer to

what happens once the controller can see everything and is optimizing the wrong thing; the present paper is what happens when the controller cannot see what it is rhetorically trying to optimize.

6.5 vs platform-moderation theoretical models

Existing theoretical work on platform moderation — Dwork et al.’s [7] window-based moderation framework, in which a platform chooses a content-moderation window and users decide whether to participate based on the fraction of admitted content aligning with their preferences — addresses domain-specific balancing under explicit policy regimes with platform-side visibility into content positions. The present paper does not claim to cover formal platform moderation: those systems have explicit policy documents that can be revised in response to errors and explicit fact-checking apparatus that constitute partial T -sensors. The mechanism named here applies to *informal* discourse regulators (academics, journalists, ad-hoc moderators, professional pundits, coalition enforcers) where no formal T -sensor exists. Dwork et al.’s framework is the natural sibling for the formal-platform case; the present paper sits where their assumed observability does not hold.

7. Worked example: local data center opposition

We work one example in detail. The example is drawn from informal discourse at the county / regional / local-coalition level, not from a formal platform-moderation case. The example is stylized: it is meant to expose the control geometry, not to adjudicate any particular project. The scope conditioning of §6.5 applies.

A county learns a large data center is coming in with tax abatements, heavy power demand, opaque procurement, and possible strain on water or grid infrastructure.

A local critic posts about sweetheart deals, infrastructure burden, contract secrecy, surveillance-contract adjacency, and long-term public cost vs private gain. Some claims are documentable; some are speculative. Nearby posters add “they’re hiding the real purpose,” “this is AI mind control,” “they’re poisoning the aquifer on purpose,” and generic 5G/UFO sludge.

What the regulatory system says it is regulating (V): truth, epistemic hygiene, anti-conspiracist discipline, keeping bad claims from spreading.

What it can actually observe quickly (y_t , which loads on V'): whether the topic is attracting cranks; whether the rhetoric resembles prior contamination genres; whether embarrassing people are piling in; whether engaging creates coalition or reputational risk; whether the discourse smells like *just asking questions*.

What it cannot observe in real time: true infrastructure burden; real subsidy structure; extent of institutional concealment; actual future harms.

What happens next. Journalists, academics, posters, and informal moderators cannot quickly resolve the object-level questions. They can quickly observe that the topic is getting weird, the discourse is aesthetically contaminated, and continued engagement carries reputational cost. Control action fires: dismissive framing, *this is how 5G/UFO nonsense starts*, policing *just asking questions* as such, treating continued curiosity as epistemic defect, steering everyone away rather than separating claims within the discourse.

The system is no longer regulating for “truth about the project.” It is regulating for distance from contamination, reputational safety, coalition legibility, avoidance of inferential slippage. *The rhetoric stays fixed* —

people still say they are protecting truth, reason, and anti-bullshit norms — but the operational target has switched.

Why the example earns its keep. The object-level question never got cleanly answered. The system did *not* establish that the project is benign, the harms are fake, the contracts fine, the concerns unfounded. It established that the topic is risky to entertain in an undifferentiated public way, that the social cost of granular inspection is too high, that topic suppression is safer than claim separation. The mechanism is visible in the outcome: the regulable variable is C , the rhetoric is fixed on T , the gap persists.

The example is ugly in exactly the right way. There are real object-level concerns; the topic also attracts spectacle and paranoia; institutions have incentives to bullshit; coalition actors are terrified of being seen as adjacent to woo. Concrete without being cartoonish.

Sibling instantiations

The same mechanism applies, with surface adjustments, to a range of regulator and operations settings. We develop one in a single paragraph and leave the others as signposts.

SRE / operations. In SRE practice, the observability asymmetry is plain: customer-visible degradation (T) often arrives through a slow, noisy channel — user reports, business metrics, postmortem reconstruction, or delayed customer-impact analysis — while dashboard SLIs (C) are cheap, high-frequency, and tightly coupled to the operator’s control loop. SLIs are indispensable, but they are not the target itself. A team can suppress alerts, improve dashboard shape, or reduce ticket inflow while leaving the underlying customer harm or substrate fragility unchanged. The pathology is not that SLIs are bad; it is that, under partial observability, the controller can begin regulating the dashboard-visible surface while continuing to describe the action as reliability work. The admissibility discipline is therefore the same: proxy evidence may authorize investigation or advisory caution, but it should not be allowed to spend authority as proof of recovered reliability.

Two further signposts, not developed here:

- **Recommender / ML systems** where the deployment objective (e.g. user well-being, long-run engagement quality) is hard to sense in real time but proxies (clicks, dwell time, immediate engagement) are cheap and fast.
- **Compliance / audit regimes** where the regulated target (e.g. genuine ethical conduct) is unsensed but proxies (paperwork completion, training-module completion rates) are cheap.

The present paper isolates the mechanism on a single primary example to keep the contribution sharp.

8. Architectural implication and open problems

The constructive consequence of Theorem 1 is essentially negative: there is no estimator- or policy-side modification, holding the same observation map fixed, that can recover T -tracking when T is genuinely unobservable. The mechanism is forced under fixed sensor geometry; the architectural escapes below all involve changing what is sensed, not how it is processed.

The architectural moves that are *not* foreclosed:

- **Restore the T -channel.** If a slow, expensive, but real T -sensor exists, install it and accept its latency. The controller can then track T at the sensor’s rate. This is the one structural escape and it is often unavailable in the cases the paper is concerned with.

- **Refuse to collapse T and V' into a single regulated variable.** Maintain separate accounting for the observable proxy and the unobservable target. Permit action under the proxy regime, but deny rhetorical closure: refuse to claim that controlling V' amounts to controlling V .
- **Witness-cohort heterogeneity.** Different agents with genuinely different $C_{\text{obs}}^{(i)}$ — for instance, witnesses positioned to observe T through different channels — *can* rotate the effective observability subspace. This does not contradict Theorem 1 within an agent; it changes the effective C_{obs} at the cohort level. Paper 24’s witness-inclusion machinery becomes relevant here as the architecture for keeping the heterogeneity intact under aggregation.

The third path is the most tractable and connects back to Paper 24 [2] and to the Governor / admissibility working track: separate sensor channels, separate admissibility accounting, separate update rules for each latent. None of these is a complete remedy; the unobservable component of T remains unobservable. What changes is whether the system rhetorically conflates the regulable V' with the nominal V , and whether the cohort can supply observation channels that the single agent cannot.

Collective observability. At the cohort level, the relevant condition is *collective observability*, which is an epistemic necessary condition: the stacked finite-horizon matrix formed from heterogeneous sensor maps $\{C_{\text{obs}}^{(i)}\}_{i=1}^N$ must not leave the target direction q in its kernel — i.e., $q \notin \ker(\bar{O}_T)$, where $\bar{O}_T$ is the finite-horizon observability matrix built from the stacked $\bar{C} = [C_{\text{obs}}^{(1)}; \dots; C_{\text{obs}}^{(N)}]$. It is not sufficient for target-tracking by itself: the architecture must also preserve the heterogeneous signal through aggregation, estimation, and control allocation; otherwise the recovered T -channel can be collapsed back into the same proxy surface the cohort was meant to escape. The fully heterogeneous case — *what cohort composition $\{C_{\text{obs}}^{(i)}\}$ restores T -observability under what aggregation rules?* — remains open and is treated in Open problems below.

Lean formalization

A companion Lean module `Paper25EpistemicBorderControl.lean` formalizes the load-bearing structural claims of §3.1 and §5:

- **Theorem 1 epistemic-access core** (`obsEquiv_policy_same`). Any policy that depends only on the observation trace is constant on observation-equivalence classes — the structural refusal of §3.1.
- **Theorem 1 corollary — target-distinct, policy-same** (`target_distinct_policy_same`). Observation-equivalent states with distinct target values still receive identical control sequences. The target-inequality hypothesis is intentionally unused in the proof; the policy never sees the target.
- **§5 kernel preservation under homogeneous witness replication** (`ker_replicateRows_eq_ker`). Stacking N homogeneous witnesses preserves the observability kernel: $\ker(\mathbf{1}_N \otimes M) = \ker(M)$ for $N > 0$.
- **§5 Gramian scaling identity** (`replicateRows_transpose_mul`). $(\mathbf{1}_N \otimes M)^\top (\mathbf{1}_N \otimes M) = N \cdot M^\top M$. Eigenspaces are invariant; eigenvalues (and squared singular values) scale by N .

The module deliberately does not formalize: (i) Diagnostic 1’s quantitative substitution scaling for general (A, B) and noise structures (paper-marked open and Lean-marked open); (ii) closed-loop Kalman / LQR dynamics (separable from the structural refusal and not needed for it); (iii) explicit finite-horizon O_T as a single matrix object (the abstract `replicateRows N M` carries the load-bearing claim); (iv) SVD or least-observable-direction quantitative claims (qualitative kernel and Gramian results are the substrate the paper actually needs). The Lean source lives in the companion repository [unpingable/lean](https://github.com/ucb-bar/unpingable/lean); the paper-side crosswalk listing module / lemma / paper-section correspondences is `docs/formalization-index.md` in the papers repository.

Open problems

1. **Quantitative version of Theorem 1.** Theorem 1’s clean statement uses identical observation trajectories as the hypothesis. A version with near-identical trajectories under noise plus a quantitative bound in terms of $|\langle v_{\min}, q \rangle|$ and $\sigma_{\min}(O_T)$ is the natural sharpening. Diagnostic 1 gives the form; promotion to a full theorem with explicit constants across (A, B, Q, R) is open.
 2. **Heterogeneous-cohort analysis.** What cohort composition $\{C_{\text{obs}}^{(i)}\}$ restores T -observability at the cohort level? When does it fail to under aggregation rules borrowed from Paper 24?
 3. **Compound-regime sim.** Does Paper 24’s witness-filter pathology stack with the present substitution mechanism, and what is the joint scaling? The algebraic adjudication of §5 establishes independence of the two mechanisms in the homogeneous case; the heterogeneous extension is where stacking would matter.
 4. **Hysteresis and asymmetric thresholds.** Threshold asymmetry in topic-classification (lower bar to enter “suspect” status than to leave) is a sibling pathology that the present model does not encode. Worth a separate treatment.
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